How Much Calibration Error Can a Price Guard Tolerate?

Fixed enforcement, lower actual product quality, unchanged welfare targets

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1 A protection promise has a model behind it

The closed recurring-cap study gives conditional customer-surplus guarantees in a known logit market. We now hold its external enforcement, physical cap, price domain and phase clock fixed, while actual inside-product quality is lower by d*mu. The outside option, buyer count and taste scale mu stay fixed. The guard still computes eligibility from its original nominal model; sellers may choose arbitrarily within that enforced set.

We keep all original absolute thresholds. The 90/95/99/100% labels refer to each old permanent-cap surplus reference, and the anchor label retains its old certified lower bound. Each requested value retains its saved 1e-8*max(1,requested value) relaxation. These targets are not reduced when true quality falls. Lower quality can reduce surplus even at unchanged prices: a failing old target is not evidence of seller gaming or extra harm caused by the controller.

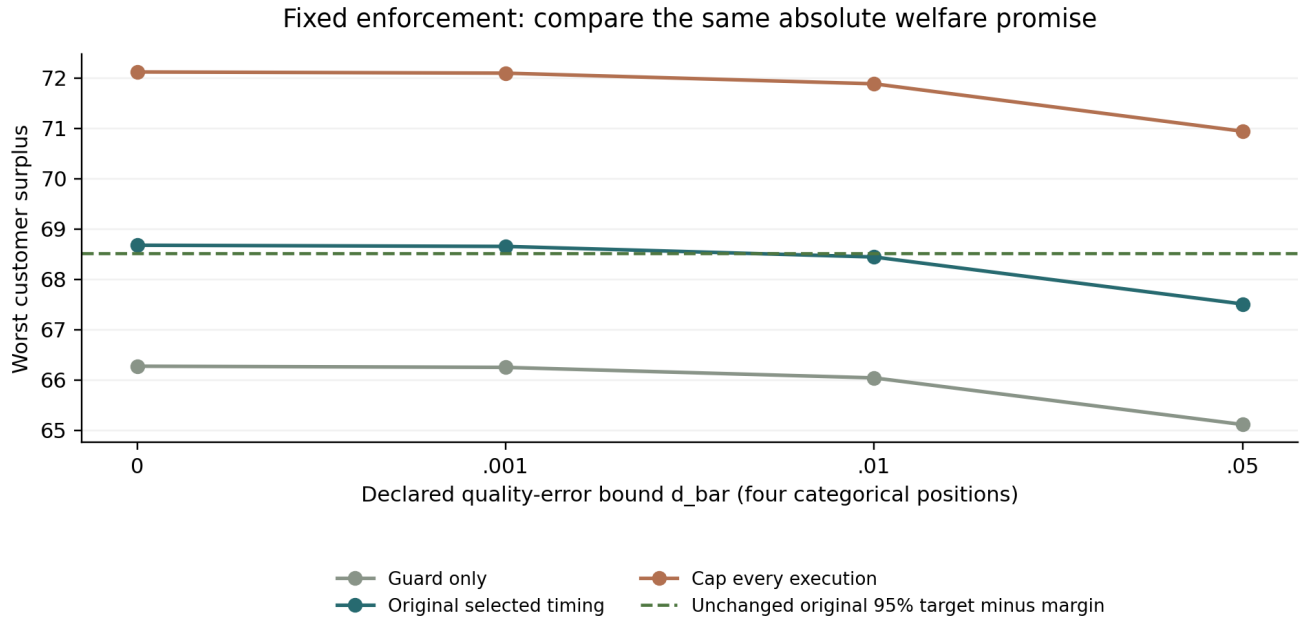


Figure1. Base effective model at four predeclared exact uncertainty bounds d_bar in {0,.001,.01,.05}. The same three policy slots and original 95%-reference threshold are compared throughout. Points are audited values; The equally spaced positions are categorical; connecting lines are visual guides, not interpolation certificates. The numerical labels are not a sample of estimated real-world errors.

Base 95% example. The already selected K=4 schedule has break-even downward error approximately 0.0069632 taste-scale units; always applying the same cap has approximately 0.153462. At the four declared bounds, the largest safe periods are 4, 4, 3, 2. The guard alone already falls below this fixed target at zero error. These are exact-model sensitivity comparisons, not a robustness claim for arbitrary demand errors.

2 The same price envelope, a different reward

Let x=(p-c)/mu and let k_hat=exp((c-g_hat)/mu), where g_hat is nominal inside quality minus outside quality and B is buyer count. Normalize the inherited physical cap as A=(p_cap-c)/mu; its previously validated premise ell<=A<q_star remains unchanged. Under the changed actual demand:

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S_hat(x,y) = [exp(-x)+exp(-y)]/k_hat
C_d(x,y) = B*mu*log[1+exp(-d)*S_hat(x,y)]
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For every finite d, C_d decreases in either price and in d. Its maximum-norm price Lipschitz constant is at most B*mu, and -B*mu<partial C_d/partial d<0. The deployed nominal upper allowance U_hat, its fixed point q_star and its contraction bound rho are unchanged. Therefore the same top-price orbit a_0=A, a_(j+1)=U_hat(a_j) dominates every permitted price pair phase by phase and attains the smallest actual surplus.

r_j(d)=C_d(a_j,a_j) still descends to C_star(d)=C_d(q_star,q_star). Thus G_K(d)=sum_(j=0..K-1)r_j(d)/K is the tight worst long-run average, and every finite average from the first real cap is at least G_K(d). Guard-only has the same all-start long-run floor C_star(d); one immediate startup cap additionally protects every finite activation average at that floor. Without this startup cap, do not assert that finite-prefix guarantee from arbitrary old prices.

For unknown fixed d in [0,d_bar], the worst endpoint is d_bar. A signed lower bound G_K(d_bar)>=F consequently protects the entire declared interval. This is not a probability statement or an empirically calibrated confidence interval. The nominal action correspondence is external and fixed; a guard recomputed from uncertain actual parameters is a different problem.

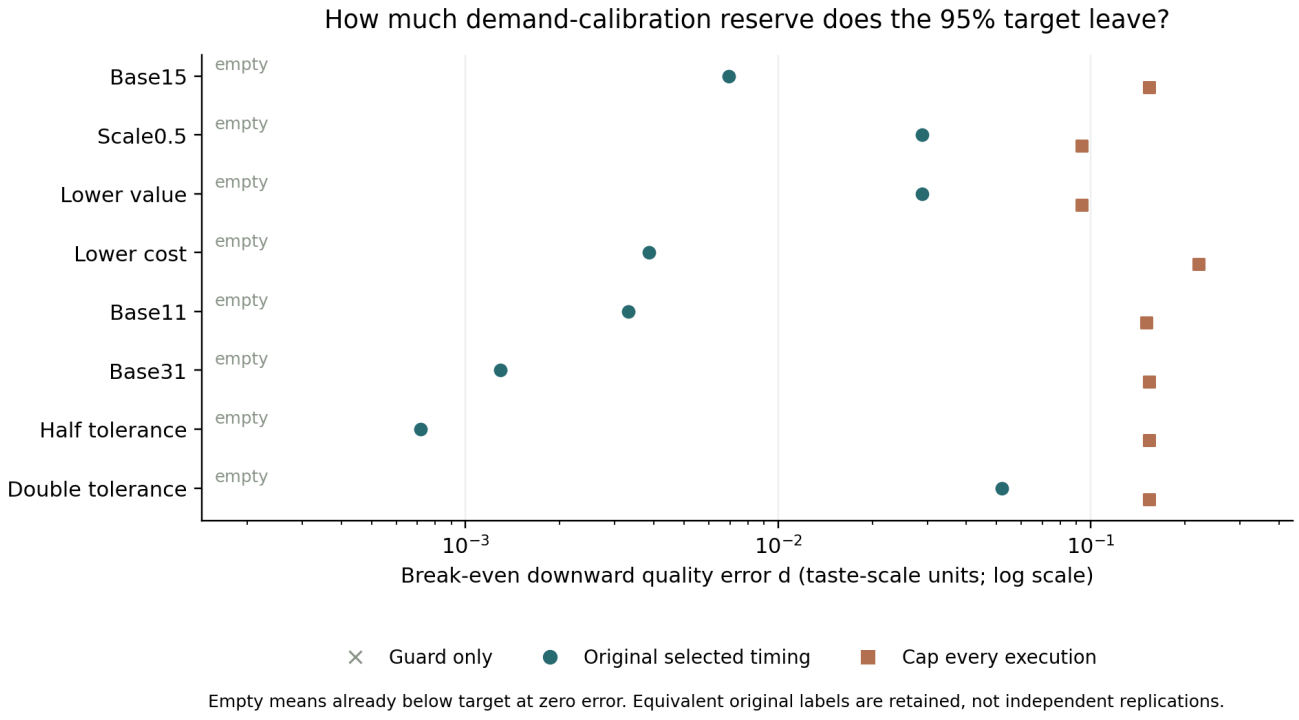
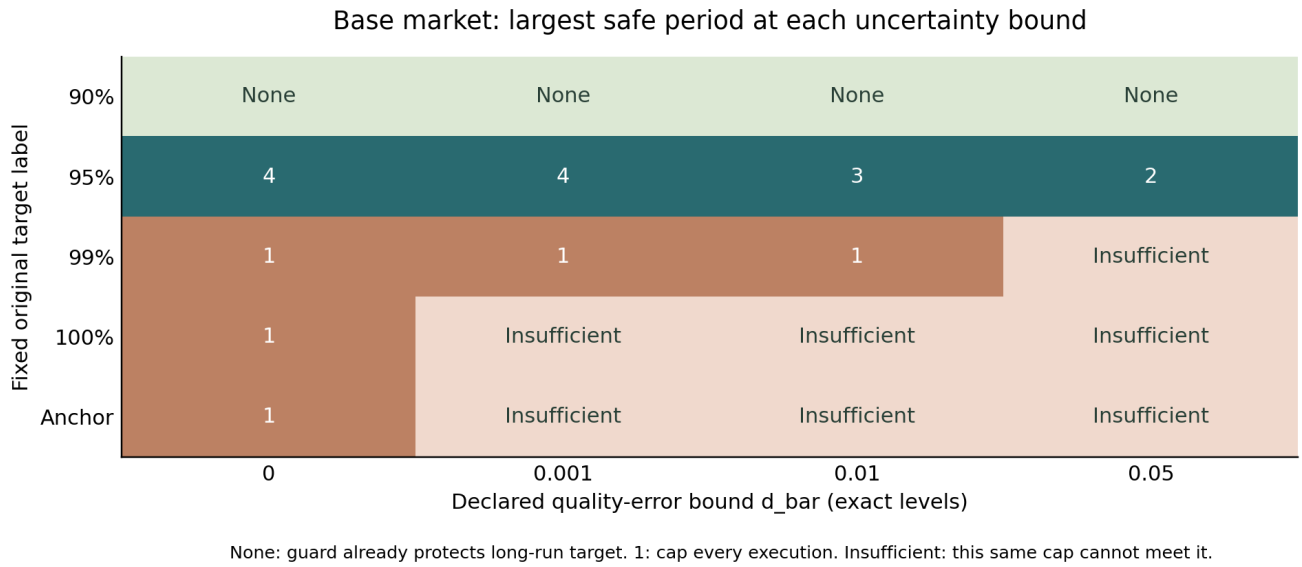


Figure2. All eight effective-model labels for the original 95% target. Positive radius endpoints are shown approximately; full signed enclosures are saved. Empty means the guard is already below target at d=0. The x axis is logarithmic and empty sets are not plotted as ordinary zero radii. Equivalent calibrations and different inherited cap/reference values are retained, not independent replications.

For F>0 and G_K(0)>F, strict decrease gives one crossing d_crit. The stationary cap/guard crossing has the separate formula log(S_hat(z,z))-log(exp(F/(B*mu))-1), with z=A or q_star. A simple sufficient radius is the positive nominal reserve divided by B*mu. Finite-period roots use signed interval endpoints; all positive crossing widths are at most 1e-20. The 64-term saved price orbit and uniform geometric price tail remain enclosed rather than replaced by invented limit rewards.

3 Stronger timing helps only while the physical cap is enough



None: guard already protects long-run target. 1: cap every execution. Insufficient: this same cap cannot meet it.

Figure3. Base effective model: largest safe integer period within the SAME physical-cap family at each predeclared bound. None concerns the long-run target; zero versus one startup cap have different finite-average scope. Insufficient means even K=1 fails this original target under the specified quality bound, not that every other controller is impossible.

The complete predeclared matrix retains 16 model semantics, 80 target groups, 240 policy-radius slots, 960 fixed-policy reserve rows and 320 robust-period comparisons. There are 54 duplicate policy slots, preserved explicitly. The independent audit certifies 174 crossing enclosures; 66 guard-policy slots already fail at zero and have EMPTY safe nonnegative sets. No observed case is unresolved or censored. All 80 zero-error period selections reproduce the closed recurring study. These are deterministic label counts, not independent experimental markets or statistical observations.

Across the 40 effective target labels, the CLOSED selected policy meets 40,26,20 and 6 targets at the four declared bounds. All 100%-reference fixed-cap targets fail already at d_bar=.001. Their approximately 1.85e-8 to 4.43e-8 taste-scale radii mostly reflect the original 1e-8 target relaxation and shrinking actual welfare. This expected reserve-budget effect is not a new emergent collusion mechanism. More frequent caps can increase reserve for some fixed targets; a near-maximal welfare promise can leave almost none.

4 Verification, limits and practical meaning

The primary sources, protocol/theory and transitive old inputs were frozen before any new alternative-demand case. The 16-model run completed in 94.38s with largest after-model sampled RSS 58.86MB; this is not a continuously measured peak. Separate 130-digit logic checked 40,598 claims across every declared row and rejected five actual-data corruptions. Primary 17 and independent 4 synthetic tests cover signs, stationary crossings, tail indexing, retained failures, period maximality and partial-output protection. The interval stacks share Python Decimal transcendental arithmetic; this is not formal verification or arithmetic-library independence.

This study concerns a one-sided common inside-quality shift under unchanged known external enforcement. It does not cover uncertain taste scale, non-logit demand, uncertain enforcement, empirically estimated error bounds, strategically stable seller behavior or autonomous discovery. The continuous-price calculations and the separately certified finite human game have distinct scopes. The graphical explainer selects only the four declared error bounds and keeps old targets fixed; no human usability result is claimed.

Sensitivity analysis and robust endpoint evaluation are established tools; no broad novelty is asserted. Closest economic context includes Brero et al.(2022), "Learning to Mitigate AI Collusion on Economic Platforms," NeurIPS 35 (ongoing intervention/robustness); Hartline, Wang and Zhang(2025), arXiv:2501.09740 (model integrity and calibrated-regret auditing, a different certificate). The contribution is a reproducible reserve comparison for this particular deployed guard/cap family, following the earlier finite-to-continuous transfer result.

Read-only audit: python research/audit-misspec-study.py. Rebuild: python research/build-misspec-figures.py; python research/build-misspec-note.py. Exact constants, frozen sources, roots, all labels, record journals and timing/memory samples remain saved. Codex agents substantially assisted formulation, code, independent AI-agent auditing, analysis and writing under Mohammadpayam Teabi's direction. No account/model game players, paid service, public deployment or submission was used in this stage. Agent review is not human peer review.